Chapter 1. Measure Spaces, Measurable Functions and Integration

σ-algebras, Dynkin systems, Lebesgue measure, measurable functions, the Lebesgue integral and its limit theorems, L^p-spaces

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1.1 Systems of sets and measures

Motivation. A probability or a length cannot be assigned consistently to *every* subset of $\mathbb{R}$; one first fixes a family of "measurable" sets that is closed under the countable operations a limit argument needs. This section introduces those families (σ-algebras, Dynkin systems, rings) and the tool that makes them useful: a measure is determined by its values on any $\cap$-stable generator.

Definition 1.1 (σ-algebra)

Let Ω be a set and $\mathcal{P}(\Omega) := \{A \mid A \subset \Omega\}$ its power set. A family $\mathcal{A} \subset \mathcal{P}(\Omega)$ is a **σ-algebra** (or σ-field) on Ω if

(i) $\Omega \in \mathcal{A}$;

(ii) $A \in \mathcal{A} \Rightarrow A^c := \Omega \setminus A \in \mathcal{A}$;

(iii) $A_i \in \mathcal{A}$, $i \in \mathbb{N} \Rightarrow \bigcup_{i\in\mathbb{N}} A_i \in \mathcal{A}$.

The prefix "σ" always signals closure under *countable* operations, as in the summation sign Σ of a series.

Example 1.2

(i) $\mathcal{P}(\Omega)$ is a σ-algebra; it is the natural choice for finite or countable Ω and in general far too large.

(ii) If $\mathcal{A}_i$, $i \in I$ (I arbitrary, possibly uncountable), are σ-algebras on Ω, then $\bigcap_{i\in I} \mathcal{A}_i = \{A \subset \Omega \mid A \in \mathcal{A}_i \ \forall i\}$ is a σ-algebra.

(iii) (*Generated σ-algebra*) For $\emptyset \neq \mathcal{A}_0 \subset \mathcal{P}(\Omega)$,

$$\sigma(\mathcal{A}_0) := \bigcap_{\mathcal{B}\ \sigma\text{-algebra},\ \mathcal{A}_0\subset\mathcal{B}} \mathcal{B}$$

is a σ-algebra by (ii) and is the smallest σ-algebra containing $\mathcal{A}_0$.

(iv) (*Borel sets*) If Ω is a topological space and $\mathcal{A}_0$ the family of open sets, $\mathcal{B}(\Omega) := \sigma(\mathcal{A}_0)$ is the **Borel σ-algebra**. Open sets, closed sets and countable unions of closed sets are Borel; nevertheless $\mathcal{B}(\mathbb{R}) \neq \mathcal{P}(\mathbb{R})$.

(v) (*Trace*) For a σ-algebra $\mathcal{A}$ on Ω and $\Omega' \subset \Omega$, $\Omega' \cap \mathcal{A} := \{A \cap \Omega' \mid A \in \mathcal{A}\}$ is a σ-algebra on Ω'. If $\Omega' \in \mathcal{A}$, then $\Omega' \cap \mathcal{A} = \{A \in \mathcal{A} \mid A \subset \Omega'\}$.

Definition 1.3 (Measure)

Let $\Omega \neq \emptyset$ and $\mathcal{A}$ a σ-algebra on Ω. A map $\mu : \mathcal{A} \to [0, \infty]$ is a **measure** on $(\Omega, \mathcal{A})$ if

(i) $\mu(\emptyset) = 0$;

(ii) (σ-additivity) $\mu(\dot{\bigcup}_{i\in\mathbb{N}} A_i) = \sum_{i=1}^{\infty} \mu(A_i)$ for pairwise disjoint $A_i \in \mathcal{A}$.

$(\Omega, \mathcal{A}, \mu)$ is a **measure space**, $(\Omega, \mathcal{A})$ a **measurable space**, and the elements of $\mathcal{A}$ are **measurable sets**. If moreover $\mu(\Omega) = 1$, μ is a **probability measure**; we then write $P := \mu$, call $(\Omega, \mathcal{A}, P)$ a **probability space** and its measurable sets **events**.

Notation. $\dot{\bigcup}_{i\in\mathbb{N}} A_i$ (or $A_1 \,\dot\cup\, A_2$) denotes a union whose members are pairwise disjoint.

Example 1.4

(i) (*Dirac measure*) For $x \in \mathbb{R}^d$, $\delta_x(A) := 1$ if $x \in A$ and 0 otherwise defines a probability measure on $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$.

(ii) (*Restriction*) If $(\Omega, \mathcal{A}, \mu)$ is a measure space and $\Omega' \in \mathcal{A}$, then $(\Omega', \mathcal{A} \cap \Omega', \mu)$ is again a measure space.

(iii) (*Counting measure*) On $(\mathbb{N}_0, \mathcal{P}(\mathbb{N}_0))$, $\mu(A) := |A| \in [0, \infty]$ (number of elements) is a measure.

Dynkin systems (λ-systems) are weaker than σ-algebras but much easier to verify; they are the standard tool for uniqueness arguments.

Definition 1.5 (Dynkin system)

$\mathcal{D} \subset \mathcal{P}(\Omega)$ is a **Dynkin system** if (i) $\Omega \in \mathcal{D}$;
(ii) $A \in \mathcal{D} \Rightarrow A^c \in \mathcal{D}$;
(iii) $A_i \in \mathcal{D}$, $i \in \mathbb{N}$, pairwise disjoint $\Rightarrow \dot{\bigcup}_i A_i \in \mathcal{D}$.

The only difference from Definition 1.1 is that (iii) is required for disjoint sequences only.

Example 1.6

(i) Every σ-algebra is a Dynkin system.

(ii) If P_1, P_2 are probability measures on $(\Omega, \mathcal{A})$, then $\mathcal{D} := \{A \in \mathcal{A} \mid P_1(A) = P_2(A)\}$ is a Dynkin system: $\Omega \in \mathcal{D}$ because both total masses are 1, complements because $P_i(A^c) = 1 - P_i(A)$, and disjoint unions by σ-additivity. (It is in general *not* a σ-algebra.)

Remark 1.7

(i) If $\mathcal{D}$ is a Dynkin system and $A, B \in \mathcal{D}$ with $A \subset B$, then $B \setminus A = (B^c \dot{\cup} A)^c \in \mathcal{D}$.

(ii) A Dynkin system that is $\cap$-stable (closed under finite intersections) is a σ-algebra.

(a) For $A, B \in \mathcal{D}$: $A \cup B = A \dot{\cup} (B \setminus (A \cap B)) \in \mathcal{D}$, by (i) and $\cap$-stability.

(b) For $A_i \in \mathcal{D}$: $\bigcup_i A_i = \dot{\bigcup}_i \left(A_i \cap \left(\bigcup_{n<i} A_n\right)^c\right)$, and each piece lies in $\mathcal{D}$ by (a), $\cap$-stability and closure under complements.

Theorem 1.8 (π–λ theorem)

Let $\mathcal{A}_0 \subset \mathcal{P}(\Omega)$ be $\cap$-stable. Then $\sigma(\mathcal{A}_0) = \mathcal{D}(\mathcal{A}_0)$, where

$$\mathcal{D}(\mathcal{A}_0) := \bigcap_{\mathcal{D} \text{ Dynkin system, } \mathcal{A}_0 \subset \mathcal{D}} \mathcal{D}$$

is the Dynkin system generated by $\mathcal{A}_0$.

Proof.

Step 1. $\sigma(\mathcal{A}_0)$ is a Dynkin system containing $\mathcal{A}_0$, so $\mathcal{D}(\mathcal{A}_0) \subset \sigma(\mathcal{A}_0)$.

Step 2. For the converse it suffices, by Remark 1.7(ii), to show that $\mathcal{D}(\mathcal{A}_0)$ is $\cap$-stable. For $D \in \mathcal{D}(\mathcal{A}_0)$ put

$$\mathcal{D}_D := \{Q \subset \Omega \mid Q \cap D \in \mathcal{D}(\mathcal{A}_0)\}.$$

This is a Dynkin system: $\Omega \cap D = D$; if $Q \cap D \in \mathcal{D}(\mathcal{A}_0)$ then $Q^c \cap D = D \setminus (Q \cap D) \in \mathcal{D}(\mathcal{A}_0)$ by Remark 1.7(i); and disjoint unions pass through $\cap D$.

Step 3. If $E \in \mathcal{A}_0$, then $\mathcal{A}_0 \subset \mathcal{D}_E$ because $\mathcal{A}_0$ is $\cap$-stable; hence $\mathcal{D}(\mathcal{A}_0) \subset \mathcal{D}_E$, i.e. $E \cap D \in \mathcal{D}(\mathcal{A}_0)$ for all $D \in \mathcal{D}(\mathcal{A}_0)$.

Step 4. Step 3 says $\mathcal{A}_0 \subset \mathcal{D}_D$ for every $D \in \mathcal{D}(\mathcal{A}_0)$, hence $\mathcal{D}(\mathcal{A}_0) \subset \mathcal{D}_D$, which is exactly the $\cap$-stability of $\mathcal{D}(\mathcal{A}_0)$. ■

Theorem 1.9 (Uniqueness of measures)

Let μ_1, μ_2 be measures on $(\Omega, \mathcal{A})$ and $\mathcal{A}_0 \subset \mathcal{A}$ a $\cap$-stable family with $\sigma(\mathcal{A}_0) = \mathcal{A}$ and $\mu_1 = \mu_2$ on $\mathcal{A}_0$. Suppose there are $\Omega_n \in \mathcal{A}_0$ with $\Omega_n \nearrow \Omega$ (i.e. $\Omega_n \subset \Omega_{n+1}$, $\bigcup_n \Omega_n = \Omega$) and $\mu_1(\Omega_n) = \mu_2(\Omega_n) < \infty$. Then $\mu_1 = \mu_2$. If μ_1, μ_2 are probability measures, the sets Ω_n are not needed.

Proof.

Step 1 (probability measures). By Example 1.6(ii), $\mathcal{D} := \{A \in \mathcal{A} \mid \mu_1(A) = \mu_2(A)\}$ is a Dynkin system containing $\mathcal{A}_0$. By Theorem 1.8, $\mathcal{A} = \sigma(\mathcal{A}_0) = \mathcal{D}(\mathcal{A}_0) \subset \mathcal{D}$, so $\mu_1 = \mu_2$.

Step 2 (localisation). In general, fix $E \in \mathcal{A}_0$ with $\mu_1(E) = \mu_2(E) < \infty$. Then $\mathcal{D}_E := \{A \in \mathcal{A} \mid \mu_1(A \cap E) = \mu_2(A \cap E)\}$ is a Dynkin system (the finiteness of $\mu_i(E)$ makes $\mu_i((A^c) \cap E) = \mu_i(E) - \mu_i(A \cap E)$ legitimate) and contains $\mathcal{A}_0$ by $\cap$-stability. Theorem 1.8 gives $\mathcal{D}_E = \mathcal{A}$.

Step 3. Apply Step 2 to $E = \Omega_n$ and use continuity from below (Theorem 1.15(ii)): $\mu_1(A) = \lim_n \mu_1(A \cap \Omega_n) = \lim_n \mu_2(A \cap \Omega_n) = \mu_2(A)$ for every $A \in \mathcal{A}$. ■

Measures are usually first defined on a small, explicit family of sets and then extended; rings are the natural such families.

Definition 1.10 (Rings, measures on rings, σ-finiteness)

(i) $\mathcal{R} \subset \mathcal{P}(\Omega)$ is a **ring** if $A, B \in \mathcal{R} \Rightarrow A \setminus B \in \mathcal{R}$ and $A \cup B \in \mathcal{R}$. Then $\emptyset \in \mathcal{R}$ and $A \cap B = A \setminus (A \setminus B) \in \mathcal{R}$, so rings are $\cap$-stable. A ring is a **σ-ring** if it is closed under countable unions; then also $\bigcap_n A_n = A_1 \setminus \bigcup_n (A_1 \setminus A_n) \in \mathcal{R}$. A ring containing Ω is an **algebra**, and a σ-ring containing Ω is a σ-algebra.

(ii) A **measure on a ring** $\mathcal{R}$ is defined as in Definition 1.3, with σ-additivity required only for disjoint $A_i \in \mathcal{R}$ whose union lies in $\mathcal{R}$.

(iii) A measure μ on a ring $\mathcal{R}$ is **σ-finite** if there are $\Omega_n \in \mathcal{R}$ with $\Omega_n \nearrow \Omega$ and $\mu(\Omega_n) < \infty$.

Theorem 1.11 (Carathéodory's extension theorem)

Let μ_0 be a σ-finite measure on a ring $\mathcal{R}$. Then there is a unique σ-finite measure μ on $\sigma(\mathcal{R})$ with $\mu = \mu_0$ on $\mathcal{R}$.

Proof [Supplement].

Step 1 (uniqueness). $\mathcal{R}$ is $\cap$-stable, and the sets $\Omega_n \in \mathcal{R}$ from σ-finiteness satisfy $\mu_1(\Omega_n) = \mu_2(\Omega_n) = \mu_0(\Omega_n) < \infty$ for any two extensions; Theorem 1.9 gives $\mu_1 = \mu_2$.

Step 2 (outer measure). For $Q \subset \Omega$ put

$$\mu^*(Q) := \inf \Big\{ \sum_{n=1}^{\infty} \mu_0(A_n) \;\Big|\; A_n \in \mathcal{R},\ Q \subset \bigcup_n A_n \Big\}.$$

Since $\Omega_n \nearrow \Omega$, covers exist. Clearly $\mu^*(\emptyset) = 0$ and μ^* is monotone. It is σ-subadditive: given $Q =$

$\bigcup_k Q_k$ with $\mu^*(Q_k) < \infty$ and $\varepsilon > 0$, choose covers $(A_{kn})_n$ of Q_k with $\sum_n \mu_0(A_{kn}) \le \mu^*(Q_k) + \varepsilon 2^{-k}$; then $(A_{kn})_{k,n}$ covers Q and $\mu^*(Q) \le \sum_k \mu^*(Q_k) + \varepsilon$.

Step 3 ($\mu^* = \mu_0$ on $\mathcal{R}$). For $A \in \mathcal{R}$, $\mu^*(A) \le \mu_0(A)$ (cover by A itself). Conversely let $A \subset \bigcup_n A_n$ with $A_n \in \mathcal{R}$. Put $B_n := A \cap A_n \setminus \bigcup_{k<n} A_k \in \mathcal{R}$; these are disjoint with union $A \in \mathcal{R}$, so σ-additivity and monotonicity of μ_0 on $\mathcal{R}$ give $\mu_0(A) = \sum_n \mu_0(B_n) \le \sum_n \mu_0(A_n)$. Hence $\mu_0(A) \le \mu^*(A)$.

Step 4 (Carathéodory-measurable sets). Let $\mathcal{M}^* := \{A \subset \Omega \mid \mu^*(Q) = \mu^*(Q \cap A) + \mu^*(Q \setminus A)\ \forall Q \subset \Omega\}$. By subadditivity only "$\ge$" must be checked, and only for $\mu^*(Q) < \infty$. $\mathcal{M}^*$ contains Ω and is closed under complements (the defining identity is symmetric in A, A^c). If $A, B \in \mathcal{M}^*$, then for every Q

$$\mu^*(Q) = \mu^*(Q \cap A) + \mu^*(Q \cap A^c \cap B) + \mu^*(Q \cap A^c \cap B^c) \ge \mu^*(Q \cap (A \cup B)) + \mu^*(Q \setminus (A \cup B)),$$

using $Q \cap (A \cup B) = (Q \cap A) \cup (Q \cap A^c \cap B)$ and subadditivity; so $\mathcal{M}^*$ is an algebra. For disjoint $A_1, \dots, A_n \in \mathcal{M}^*$ induction gives $\mu^*(Q \cap \bigcup_{i \le n} A_i) = \sum_{i \le n} \mu^*(Q \cap A_i)$. Now let $A_i \in \mathcal{M}^*$ be disjoint, $A := \bigcup_i A_i$, $S_n := \bigcup_{i \le n} A_i \in \mathcal{M}^*$. Then

$$\mu^*(Q) = \mu^*(Q \cap S_n) + \mu^*(Q \setminus S_n) \ge \sum_{i \le n} \mu^*(Q \cap A_i) + \mu^*(Q \setminus A),$$

and letting $n \to \infty$ and using σ-subadditivity, $\mu^*(Q) \ge \sum_i \mu^*(Q \cap A_i) + \mu^*(Q \setminus A) \ge \mu^*(Q \cap A) + \mu^*(Q \setminus A)$. Thus $A \in \mathcal{M}^*$; since an algebra closed under disjoint countable unions is closed under all countable unions (disjointify as in Remark 1.7(ii)(b)), $\mathcal{M}^*$ is a σ-algebra. Taking $Q = A$ in the last display shows that μ^* is σ-additive on $\mathcal{M}^*$.

Step 5 ($\mathcal{R} \subset \mathcal{M}^*$). Let $A \in \mathcal{R}$, $Q \subset \Omega$ with $\mu^*(Q) < \infty$, and let $(A_n) \subset \mathcal{R}$ cover Q with $\sum \mu_0(A_n) \le \mu^*(Q) + \varepsilon$. Then $(A_n \cap A)$ covers $Q \cap A$, $(A_n \setminus A)$ covers $Q \setminus A$, and $\mu_0(A_n) = \mu_0(A_n \cap A) + \mu_0(A_n \setminus A)$, so $\mu^*(Q \cap A) + \mu^*(Q \setminus A) \le \mu^*(Q) + \varepsilon$.

Step 6. By Steps 4–5, $\sigma(\mathcal{R}) \subset \mathcal{M}^*$, and $\mu := \mu^*|_{\sigma(\mathcal{R})}$ is a measure extending μ_0 (Step 3). It is σ-finite because $\mu(\Omega_n) = \mu_0(\Omega_n) < \infty$. ∎

Remark 1.12

In the lectures the extension theorem is not used for Lebesgue measure. Instead, Section 1.2 constructs it directly, following Rudin's *Principles of Mathematical Analysis*, Ch. 11. The idea is the same outer-measure argument as above, but the abstract ring is replaced by the concrete ring of elementary sets in $\mathbb{R}^d$, so that compactness and open/closed approximation can be used. Rudin does not first prove σ-additivity on elementary sets: he introduces the outer measure at once, defines measurability by approximation with elementary sets, and obtains σ-additivity only on the resulting σ-ring. Much of the literature starts from the ring of half-open intervals instead; elementary sets give a shorter route here.

1.2 Construction of the Lebesgue measure

Motivation. We want a measure λ on $\mathbb{R}^d$ that gives a box its volume. Volume is obviously additive on finite unions of boxes; the work is to extend it to a σ-additive set function on a σ-algebra containing all open sets. The route is: outer measure → a pseudo-metric on sets → the sets that can be approximated by elementary sets.

Definition 1.13 (Additive and σ-additive set functions)

Let $\varphi : \mathcal{R} \to [0,\infty]$ on a ring $\mathcal{R}$ with $\varphi(\emptyset) = 0$. φ is **additive** if

$$A, B \in \mathcal{R},\ A \cap B = \emptyset \implies \varphi(A \cup B) = \varphi(A) + \varphi(B), \tag{1.1}$$

and **σ-additive** (a measure on $\mathcal{R}$) if $\varphi(\dot{\bigcup}_i A_i) = \sum_i \varphi(A_i)$ whenever the $A_i \in \mathcal{R}$ are pairwise disjoint and $\bigcup_i A_i \in \mathcal{R}$.

Remark 1.14

If φ is additive on $\mathcal{R}$, then by induction it is finitely additive, $\varphi(A_1 \dot{\cup} \cdots \dot{\cup} A_n) = \sum_i \varphi(A_i)$, and for all $A, B \in \mathcal{R}$:

$$\varphi(A \cup B) + \varphi(A \cap B) = \varphi(A) + \varphi(B), \tag{1.2}$$

$$A \subset B \Rightarrow \varphi(A) \le \varphi(B) \quad \text{(monotonicity)}, \tag{1.3}$$

$$A \subset B,\ \varphi(A) < \infty \Rightarrow \varphi(B \setminus A) = \varphi(B) - \varphi(A),$$

$$\varphi\Big(\bigcup_{i\le n} A_i\Big) \le \sum_{i\le n} \varphi(A_i) \quad \text{(finite subadditivity)}.$$

For (1.2) write $A \cup B = (A\setminus B) \dot{\cup} (A\cap B) \dot{\cup} (B\setminus A)$ and $A = (A\setminus B) \dot{\cup} (A\cap B)$, $B = (B\setminus A) \dot{\cup} (A\cap B)$. Monotonicity is $\varphi(B) = \varphi(A) + \varphi(B \setminus A)$, and subadditivity follows from (1.2) and induction.

Notation. $A_n \nearrow A$ means $A_n \subset A_{n+1}$ and $A = \bigcup_n A_n$; $A_n \searrow A$ means $A_n \supset A_{n+1}$ and $A = \bigcap_n A_n$.

Theorem 1.15 (Continuity of measures)

Let φ be additive on a ring $\mathcal{R}$ and consider

(i) φ is σ-additive on $\mathcal{R}$;

(ii) (continuity from below) $A_n \in \mathcal{R}$, $A_n \nearrow A \in \mathcal{R} \Rightarrow \varphi(A) = \lim_n \varphi(A_n)$;

(iii) (continuity from above) $A_n \in \mathcal{R}$, $A_n \searrow A \in \mathcal{R}$, $\varphi(A_{n_0}) < \infty$ for some $n_0 \Rightarrow \varphi(A) = \lim_n \varphi(A_n)$.

Then (i) $\Leftrightarrow$ (ii) $\Rightarrow$ (iii).

Proof.

Step (i)$\Rightarrow$(ii). Put $B_1 := A_1$, $B_n := A_n \setminus A_{n-1}$. The $B_n \in \mathcal{R}$ are disjoint, $A_n = B_1 \dot{\cup} \cdots \dot{\cup} B_n$ and $\bigcup_n B_n = A$, so $\lim_n \varphi(A_n) = \lim_n \sum_{i\le n} \varphi(B_i) = \sum_i \varphi(B_i) = \varphi(A)$.

Step (ii)$\Rightarrow$(i). For disjoint $A_n \in \mathcal{R}$ with $A := \bigcup_n A_n \in \mathcal{R}$, $B_N := \bigcup_{n\le N} A_n \nearrow A$, so $\varphi(A) = \lim_N \varphi(B_N) = \lim_N \sum_{n\le N} \varphi(A_n)$.

Step (ii)$\Rightarrow$(iii) [Supplement]. For $n \ge n_0$ the sets $C_n := A_{n_0} \setminus A_n \in \mathcal{R}$ increase to $A_{n_0} \setminus A \in \mathcal{R}$. All quantities are finite because $\varphi(A_n), \varphi(A) \le \varphi(A_{n_0}) < \infty$, so by Remark 1.14 and (ii)

$$\varphi(A_{n_0}) - \varphi(A) = \varphi(A_{n_0} \setminus A) = \lim_n \varphi(C_n) = \varphi(A_{n_0}) - \lim_n \varphi(A_n).$$

■

Definition 1.16 (Intervals and elementary sets)

For $a, b \in \mathbb{R}^d$ with $a_i \le b_i$, an **interval** (bounded box) is any set of points x with $a_i \lhd_i x_i \lhd_i' b_i$, $\lhd_i, \lhd_i' \in \{\le, <\}$. It is open (closed) if all inequalities are strict (non-strict); $a_i = b_i$ is allowed,

so an interval may be empty or a point. An **elementary set** is a finite union of intervals; $\mathcal{E}$ denotes the family of elementary sets. For an interval I put $\lambda(I) := \prod_{i=1}^{d}(b_i - a_i)$, and for $A = I_1 \dot\cup \cdots \dot\cup I_n$ with disjoint intervals

$$\lambda(A) := \sum_{i=1}^{n} \lambda(I_i). \tag{1.4}$$

For $d = 1, 2, 3$, λ is length, area and volume. The following facts are elementary. $\mathcal{E}$ is a ring but not a σ-ring: for $d = 1$, $\mathbb{Q} \cap [0, 1]$ is a countable union of one-point intervals but not a finite union of intervals. Every $A \in \mathcal{E}$ is a finite *disjoint* union of intervals: the endpoints of the intervals in all coordinates define a grid whose cells are intervals, and A is a union of cells. And $\lambda(A)$ in (1.4) does not depend on the decomposition: two decompositions have a common refinement into grid cells, and λ is additive on the cells of a grid inside one box. Hence λ is a finite, additive, nonnegative set function on $\mathcal{E}$.

Definition 1.17 (Regularity)

An additive $\varphi : \mathcal{E} \to [0, \infty]$ is **regular** if for every $A \in \mathcal{E}$ and $\varepsilon > 0$ there are $F, G \in \mathcal{E}$, F closed, G open, with $F \subset A \subset G$ and $\varphi(G) - \varepsilon \le \varphi(A) \le \varphi(F) + \varepsilon$.

Example 1.18

λ is regular and finite on $\mathcal{E}$: an interval can be shrunk to a closed one and enlarged to an open one, changing the product of side lengths by less than ε, and an elementary set is a finite disjoint union of intervals.

From now on μ denotes a finite, regular, additive set function on $\mathcal{E}$ (for example λ). The goal is to extend it to a σ-additive set function on a σ-ring containing $\mathcal{E}$.

Definition 1.19 (Outer measure)

For $E \subset \mathbb{R}^d$,

$$\mu^*(E) := \inf\Big\{ \sum_{n=1}^{\infty} \mu(A_n) \;\Big|\; A_n \in \mathcal{E} \text{ open},\ E \subset \bigcup_n A_n \Big\}.$$

Clearly $\mu^*(E) \ge 0$, $\mu^*(\emptyset) = 0$, and $E_1 \subset E_2 \Rightarrow \mu^*(E_1) \le \mu^*(E_2)$.

Theorem 1.20

(i) $\mu^*(A) = \mu(A)$ for $A \in \mathcal{E}$, so μ^* extends μ to $\mathcal{P}(\mathbb{R}^d)$.

(ii) (σ-subadditivity) If $E = \bigcup_n E_n$ with arbitrary $E_n \subset \mathbb{R}^d$, then

$$\mu^*(E) \le \sum_{n=1}^{\infty} \mu^*(E_n). \tag{1.5}$$

Proof. (i) Let $A \in \mathcal{E}$, $\varepsilon > 0$. By regularity there is an open $G \in \mathcal{E}$ with $G \supset A$ and $\mu(G) \le \mu(A) + \varepsilon$; G alone is an admissible cover, so $\mu^*(A) \le \mu(A)$.

Conversely, take open $A_n \in \mathcal{E}$ covering A with $\sum_n \mu(A_n) \le \mu^*(A) + \varepsilon$, and a closed $F \in \mathcal{E}$, $F \subset A$, with $\mu(A) \le \mu(F) + \varepsilon$. F is closed and bounded, hence compact, so $F \subset A_1 \cup \cdots \cup A_N$ for some N. By monotonicity and finite subadditivity,

$$\mu(A) \le \mu(F) + \varepsilon \le \mu(A_1 \cup \cdots \cup A_N) + \varepsilon \le \sum_{k \le N} \mu(A_k) + \varepsilon \le \mu^*(A) + 2\varepsilon.$$

(ii) Assume $\mu^*(E_n) < \infty$ for all n (otherwise there is nothing to show). For $\varepsilon > 0$ choose open covers $(A_{nk})_k \subset \mathcal{E}$ of E_n with $\sum_k \mu(A_{nk}) \le \mu^*(E_n) + 2^{-n}\varepsilon$. Then $(A_{nk})_{n,k}$ is an open cover of E, so $\mu^*(E) \le \sum_n \sum_k \mu(A_{nk}) \le \sum_n \mu^*(E_n) + \varepsilon$. ■

Definition 1.21 (Approximation by elementary sets)

For $A, B \subset \mathbb{R}^d$ let

$$S(A,B) := (A \setminus B) \cup (B \setminus A) \quad \text{(symmetric difference)}, \qquad d(A,B) := \mu^*(S(A,B)) \in [0,\infty].$$

Write $A_n \to A$ if $d(A, A_n) \to 0$, and define

$$A \in \mathcal{M}_F(\mu) \text{ (finitely } \mu\text{-measurable)} :\Longleftrightarrow \exists A_n \in \mathcal{E} : \ A_n \to A,$$
$$A \in \mathcal{M}(\mu) \text{ } (\mu\text{-measurable)} :\Longleftrightarrow \ A = \bigcup_n A_n \text{ with } A_n \in \mathcal{M}_F(\mu).$$

Obviously $\mathcal{E} \subset \mathcal{M}_F(\mu) \subset \mathcal{M}(\mu)$. Every $A \in \mathcal{M}_F(\mu)$ has $\mu^*(A) < \infty$: for n large, $A \subset S(A, A_n) \cup A_n$, so $\mu^*(A) \le d(A, A_n) + \mu(A_n) < \infty$.

Theorem 1.22

$\mathcal{M}(\mu)$ is a σ-ring and μ^* is σ-additive on $\mathcal{M}(\mu)$.

The proof needs two lemmas.

Lemma 1.23

For $A, B, C \subset \mathbb{R}^d$:

(1) $S(A,B) = S(B,A)$;

(2) $S(A,B) \subset S(A,C) \cup S(C,B)$;

(3) $S(A,B) = S(A^c, B^c)$;

(4) for A_1, A_2, B_1, B_2, each of $S(A_1 \cup A_2, B_1 \cup B_2)$, $S(A_1 \cap A_2, B_1 \cap B_2)$, $S(A_1 \setminus A_2, B_1 \setminus B_2)$ is contained in $S(A_1, B_1) \cup S(A_2, B_2)$.

Consequently, by monotonicity and subadditivity of μ^*: (i) $d(A,B) = d(B,A)$;
(ii) $d(A,B) \le d(A,C) + d(C,B)$;
(iii) $A = B \Rightarrow d(A,B) = 0$ (not conversely);
(iv) $d(A_1 \star A_2, B_1 \star B_2) \le d(A_1, B_1) + d(A_2, B_2)$ for $\star \in \{\cup, \cap, \setminus\}$.

Proof. (1) is clear.

(2) If $x \in A \setminus B$, then either $x \in C$ (so $x \in C \setminus B$) or $x \notin C$ (so $x \in A \setminus C$); hence $A \setminus B \subset (A \setminus C) \cup (C \setminus B)$, and symmetrically $B \setminus A \subset (B \setminus C) \cup (C \setminus A)$. The union of the right-hand sides is $S(A,C) \cup S(C,B)$.

(3) $S(A,B) = (A \cap B^c) \cup (B \cap A^c) = (B^c \setminus A^c) \cup (A^c \setminus B^c) = S(A^c, B^c)$.

(4) Since $(A_1 \cup A_2) \setminus (B_1 \cup B_2) \subset (A_1 \setminus B_1) \cup (A_2 \setminus B_2)$ and symmetrically,

$$S(A_1 \cup A_2, B_1 \cup B_2) \subset S(A_1, B_1) \cup S(A_2, B_2). \tag{1.6}$$

For intersections use (3) and (1.6): $S(A_1 \cap A_2, B_1 \cap B_2) = S(A_1^c \cup A_2^c, B_1^c \cup B_2^c) \subset S(A_1^c, B_1^c) \cup S(A_2^c, B_2^c) = S(A_1, B_1) \cup S(A_2, B_2)$. For differences write $A_1 \setminus A_2 = A_1 \cap A_2^c$ and use the intersection case together with (3). ■

Lemma 1.24

If $\mu^*(A) < \infty$ or $\mu^*(B) < \infty$, then

$$|\mu^*(A) - \mu^*(B)| \le d(A, B). \tag{1.7}$$

Proof. Say $\mu^*(B) \le \mu^*(A)$, so $\mu^*(B) < \infty$. Then by Lemma 1.23(ii),

$$\mu^*(A) = d(A, \emptyset) \le d(A, B) + d(B, \emptyset) = d(A, B) + \mu^*(B).$$

The case $\mu^*(A) \le \mu^*(B)$ is symmetric. ∎

Remark 1.25

(i) For $\mu = \lambda$, $d = 1$, a countable $A \subset \mathbb{R}$ and $B = \emptyset$: $d(A, B) = \lambda^*(A) \le \sum_{p \in A} \lambda([p, p]) = 0$, although $A \ne B$. So d is only a pseudo-metric; A and B differ by a λ^*-null set.

(ii) On $\mathcal{F} := \{A \subset \mathbb{R}^d \mid \mu^*(A) < \infty\} \supset \mathcal{M}_F(\mu)$ identify A and B when $d(A, B) = 0$. Then $(\mathcal{F}, d)$ is a metric space and $\mathcal{M}_F(\mu)$ is the closure $\overline{\mathcal{E}}$ of $\mathcal{E}$ in it. [Supplement] *Reason:* d is finite on $\mathcal{F}$ by subadditivity and is symmetric with the triangle inequality by Lemma 1.23. $\mathcal{M}_F(\mu) \subset \overline{\mathcal{E}}$ is the definition. For the reverse inclusion, a limit of points of $\overline{\mathcal{E}}$ is a limit of elementary sets by a diagonal choice and the triangle inequality.

Proof of Theorem 1.22.

Step 1 ($\mathcal{M}_F(\mu)$ is a ring). Let $A, B \in \mathcal{M}_F(\mu)$ with elementary $A_n \to A$, $B_n \to B$. By Lemmas 1.23(iv) and 1.24,

$$\begin{aligned} &d(A_n \cup B_n, A \cup B) + d(A_n \cap B_n, A \cap B) + d(A_n \setminus B_n, A \setminus B) + |\mu^*(A_n) - \mu^*(A)| + |\mu^*(B_n) - \mu^*(B)| \\ &\quad + |\mu^*(A_n \cup B_n) - \mu^*(A \cup B)| + |\mu^*(A_n \cap B_n) - \mu^*(A \cap B)| \le 6\big(d(A_n, A) + d(B_n, B)\big) \to 0. \end{aligned}$$

Since $A_n \cup B_n$, $A_n \setminus B_n \in \mathcal{E}$, we get $A \cup B$, $A \setminus B \in \mathcal{M}_F(\mu)$, and also

$$\mu^*(A_n) \to \mu^*(A), \quad \mu^*(A_n \cup B_n) \to \mu^*(A \cup B), \quad \mu^*(A_n \cap B_n) \to \mu^*(A \cap B). \tag{1.8}$$

Step 2 (μ^* is additive on $\mathcal{M}_F(\mu)$). By (1.2) for $\mu = \mu^*$ on $\mathcal{E}$, $\mu^*(A_n) + \mu^*(B_n) = \mu^*(A_n \cup B_n) + \mu^*(A_n \cap B_n)$. Letting $n \to \infty$ with (1.8): $\mu^*(A) + \mu^*(B) = \mu^*(A \cup B) + \mu^*(A \cap B)$, and $\mu^*(A \cap B) = 0$ if $A \cap B = \emptyset$.

Step 3 (disjoint representation). Every $A \in \mathcal{M}(\mu)$ is a disjoint union of sets from $\mathcal{M}_F(\mu)$: if $A = \bigcup_n A'_n$ with $A'_n \in \mathcal{M}_F(\mu)$, put $B'_n := A'_1 \cup \cdots \cup A'_n$, $A_1 := B'_1$, $A_n := B'_n \setminus B'_{n-1}$ (all in the ring $\mathcal{M}_F(\mu)$). For such a disjoint representation, σ-subadditivity gives $\mu^*(A) \le \sum_n \mu^*(A_n)$, while additivity and monotonicity give $\sum_{k \le n} \mu^*(A_k) = \mu^*(A_1 \dot\cup \cdots \dot\cup A_n) \le \mu^*(A)$. Hence

$$\mu^*(A) = \sum_{n=1}^{\infty} \mu^*(A_n). \tag{1.9}$$

Step 4 (finite measure implies $\mathcal{M}_F$). Let $A \in \mathcal{M}(\mu)$ with $\mu^*(A) < \infty$, written as in Step 3, and $B_n := A_1 \cup \cdots \cup A_n \in \mathcal{M}_F(\mu)$. Then $d(A, B_n) = \mu^*\big(\bigcup_{k>n} A_k\big) = \sum_{k>n} \mu^*(A_k) \to 0$ by (1.9) (Step 3 applied to $\bigcup_{k>n} A_k$). Choose elementary B_{n,k_n} with $d(B_{n,k_n}, B_n) < 1/n$; by the triangle inequality $B_{n,k_n} \to A$. Thus

$$A \in \mathcal{M}(\mu),\ \mu^*(A) < \infty \implies A \in \mathcal{M}_F(\mu). \tag{1.10}$$

Step 5 (σ-additivity on $\mathcal{M}(\mu)$). Let $A_n \in \mathcal{M}(\mu)$ be disjoint with union A. If $\mu^*(A) = \infty$, σ-subadditivity forces $\sum_n \mu^*(A_n) = \infty$. If $\mu^*(A) < \infty$, every A_n has finite outer measure, so $A_n \in \mathcal{M}_F(\mu)$ by (1.10), and (1.9) gives $\mu^*(A) = \sum_n \mu^*(A_n)$.

Step 6 ($\mathcal{M}(\mu)$ is a σ-ring). If $A_n = \bigcup_k A_{nk}$ with $A_{nk} \in \mathcal{M}_F(\mu)$, then $\bigcup_n A_n = \bigcup_{(n,k)\in\mathbb{N}^2} A_{nk} \in \mathcal{M}(\mu)$. For differences let $A = \bigcup_n A_n$, $B = \bigcup_l B_l$ with $A_n, B_l \in \mathcal{M}_F(\mu)$. Then $A_n \cap B = \bigcup_l (A_n \cap B_l) \in \mathcal{M}(\mu)$ has $\mu^*(A_n \cap B) \le \mu^*(A_n) < \infty$, so $A_n \cap B \in \mathcal{M}_F(\mu)$ by (1.10), and $A_n \setminus B = A_n \setminus (A_n \cap B) \in \mathcal{M}_F(\mu)$. Hence $A \setminus B = \bigcup_n (A_n \setminus B) \in \mathcal{M}(\mu)$. ■

Definition 1.26 (Lebesgue measure)

Set $\mu(A) := \mu^*(A)$ for $A \in \mathcal{M}(\mu)$. This extends μ from $\mathcal{E}$ to a σ-additive set function on the σ-ring $\mathcal{M}(\mu)$, i.e. a measure in the sense of Definition 1.10(ii). For $\mu = \lambda$ it is the **Lebesgue measure** on $\mathbb{R}^d$.

Remark 1.27

(i) $\mathcal{M}(\mu)$ contains the open sets, hence $\mathbb{R}^d$, so it is a σ-algebra and contains the closed sets too. Indeed, every open ball is a countable union of open intervals, and every open set is a countable union of balls with rational centres and radii; hence every open set is a countable union of open intervals, which lie in $\mathcal{E} \subset \mathcal{M}(\mu)$.

(ii) (*Regularity*) For $A \in \mathcal{M}(\mu)$ and $\varepsilon > 0$ there are F closed and G open with $F \subset A \subset G$, $\mu(G \setminus A) < \varepsilon$ and $\mu(A \setminus F) < \varepsilon$. The set G exists by Proposition 1.28. For F, apply Proposition 1.28 to A^c: there is an open $\tilde{G} \supset A^c$ with $\mu(\tilde{G} \setminus A^c) < \varepsilon$, and $F := \tilde{G}^c \subset A$ is closed with $A \setminus F = A \cap \tilde{G} = \tilde{G} \setminus A^c$.

(iii) With $G_n \supset A \supset F_n$ from (ii) for $\varepsilon = 1/n$, monotonicity gives $\mu(\bigcap_n G_n \setminus A) = 0 = \mu(A \setminus \bigcup_n F_n)$. Thus $A = B \dot\cup (A \setminus B)$ with $B := \bigcup_n F_n \in \mathcal{B}(\mathbb{R}^d)$ and $A \setminus B$ a μ-null set. Every μ-measurable set is a Borel set plus a null set, and always $\mathcal{B}(\mathbb{R}^d) \subset \mathcal{M}(\mu)$.

(iv) The μ-null sets form a σ-ring, and every $N \subset \mathbb{R}^d$ with $\mu^*(N) = 0$ is a μ-null set. [Supplement] *Reason:* countable unions of null sets are null by σ-subadditivity and differences by monotonicity; if $\mu^*(N) = 0$ then $d(N, \emptyset) = 0$ with $\emptyset \in \mathcal{E}$, so $N \in \mathcal{M}_F(\mu)$.

(v) For λ every countable set is a null set, but there are uncountable null sets, e.g. the Cantor set.

Proposition 1.28

μ is regular on $\mathcal{M}(\mu)$: for $A \in \mathcal{M}(\mu)$ and $\varepsilon > 0$ there is an open $G \supset A$ with $\mu(G \setminus A) < \varepsilon$ (and hence Remark 1.27(ii) holds).

Proof.

Step 1 ($\mu(A) < \infty$). Then $A \in \mathcal{M}_F(\mu)$ by (1.10), and there are open $O_n \in \mathcal{E}$ with $\bigcup_n O_n \supset A$ and $\sum_n \mu(O_n) < \mu(A) + \varepsilon$. $G := \bigcup_n O_n$ is open and $\mu(G) \le \sum_n \mu(O_n) < \mu(A) + \varepsilon$. Since $\mu(A) < \infty$, additivity gives $\mu(G \setminus A) = \mu(G) - \mu(A) < \varepsilon$.

Step 2 ($\mu(A) = \infty$) [Supplement]. Write $A = \bigcup_n A_n$ with $A_n \in \mathcal{M}_F(\mu)$ (so $\mu(A_n) < \infty$). By Step 1 choose open $G_n \supset A_n$ with $\mu(G_n \setminus A_n) < \varepsilon 2^{-n}$. Then $G := \bigcup_n G_n \supset A$ is open and $G \setminus A \subset \bigcup_n (G_n \setminus A_n)$, so $\mu(G \setminus A) < \varepsilon$. ■

1.3 Measure spaces and measurable functions

Motivation. Integration theory is no simpler on $\mathbb{R}^d$ than on an abstract set, so from now on $(X, \mathcal{M}, \mu)$ is an arbitrary measure space. Before integrating we need the functions whose level sets

are measurable, and we need to know that this class is stable under the operations used in analysis: limits, sums, products, continuous transformations.

Example 1.29

(i) $(\mathbb{R}^d, \mathcal{M}(\lambda), \lambda)$ is the **Lebesgue measure space**.
(ii) $\lambda|_{\mathcal{B}(\mathbb{R}^d)}$ is the **Lebesgue–Borel measure**, and $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d), \lambda)$ the Lebesgue–Borel measure space; in practice it is also simply called Lebesgue measure.

Definition 1.30 (Measurable function)

An extended real-valued $f : X \to \overline{\mathbb{R}} = [-\infty, \infty]$ is **measurable** if $\{f > \alpha\} := \{x \in X \mid f(x) \in (\alpha, \infty]\} \in \mathcal{M}$ for every $\alpha \in \mathbb{R}$.

This depends only on f and $\mathcal{M}$: for $(\mathbb{R}^d, \mathcal{M}(\lambda))$ one says *Lebesgue measurable*, for $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$ *Borel measurable*.

Example 1.31

(i) Borel measurable functions are Lebesgue measurable. The converse fails: $\mathcal{B}(\mathbb{R}^d)$ has the cardinality $\mathfrak{c}$ of the continuum while $\mathcal{M}(\lambda)$ has cardinality $2^{\mathfrak{c}}$ (Dudley, *Real Analysis and Probability*, Prop. 3.4.1), so there is a Lebesgue measurable, non-Borel set L, and $\mathbb{1}_L$ is Lebesgue but not Borel measurable. (ii) A continuous $f : \mathbb{R}^d \to \mathbb{R}$ is Borel measurable, since $\{f > \alpha\} = f^{-1}((\alpha, \infty))$ is open.

Remark 1.32

For measurable spaces $(X, \mathcal{M})$, $(\tilde{X}, \tilde{\mathcal{M}})$, a map $T : X \to \tilde{X}$ is $\mathcal{M}/\tilde{\mathcal{M}}$**-measurable** if $T^{-1}(\tilde{A}) \in \mathcal{M}$ for all $\tilde{A} \in \tilde{\mathcal{M}}$. If $\tilde{\mathcal{M}} = \sigma(\tilde{\mathcal{M}}_0)$, it suffices that $T^{-1}(\tilde{A}) \in \mathcal{M}$ for all $\tilde{A} \in \tilde{\mathcal{M}}_0$. [Supplement] *Reason:* $\{\tilde{A} \subset \tilde{X} \mid T^{-1}(\tilde{A}) \in \mathcal{M}\}$ is a σ-algebra, since preimages commute with complements and unions, and it contains $\tilde{\mathcal{M}}_0$, hence $\sigma(\tilde{\mathcal{M}}_0)$. In particular, with $\mathcal{B}(\overline{\mathbb{R}}) := \{A \subset \overline{\mathbb{R}} \mid A \cap \mathbb{R} \in \mathcal{B}(\mathbb{R})\} = \sigma(\{(\alpha, \infty] \mid \alpha \in \mathbb{R}\})$, Definition 1.30 is exactly $\mathcal{M}/\mathcal{B}(\overline{\mathbb{R}})$-measurability.

Theorem 1.33

For $f : X \to \overline{\mathbb{R}}$ the following are equivalent: (i) $\{f > \alpha\} \in \mathcal{M}\ \forall \alpha \in \mathbb{R}$;
(ii) $\{f \geq \alpha\} \in \mathcal{M}\ \forall \alpha$;
(iii) $\{f < \alpha\} \in \mathcal{M}\ \forall \alpha$;
(iv) $\{f \leq \alpha\} \in \mathcal{M}\ \forall \alpha$.

Proof. For $\alpha \in \mathbb{R}$:

(i)⇒(ii): $\{f \geq \alpha\} = \bigcap_n \{f > \alpha - \frac{1}{n}\}$.

(ii)⇒(iii): $\{f < \alpha\} = X \setminus \{f \geq \alpha\}$.

(iii)⇒(iv): $\{f \leq \alpha\} = \bigcap_n \{f < \alpha + \frac{1}{n}\}$.

(iv)⇒(i): $\{f > \alpha\} = X \setminus \{f \leq \alpha\}$. ∎

Theorem 1.34

f measurable $\Rightarrow |f|$ measurable.

Proof. $\{|f| < \alpha\} = \{f < \alpha\} \cap \{f > -\alpha\} \in \mathcal{M}$; apply Theorem 1.33. ■

Theorem 1.35

If f_n, $n \in \mathbb{N}$, are measurable, so are $\inf_n f_n$, $\sup_n f_n$, $\liminf_n f_n$ and $\limsup_n f_n$.

Proof. $\{\sup_n f_n > \alpha\} = \bigcup_n \{f_n > \alpha\}$ and $\{\inf_n f_n < \alpha\} = \bigcup_n \{f_n < \alpha\}$. Then $\limsup_n f_n = \inf_n \sup_{m \geq n} f_m$ and $\liminf_n f_n = \sup_n \inf_{m \geq n} f_m$ are measurable by two applications. ■

Corollary 1.36

(i) f, g measurable $\Rightarrow$ $\max(f, g)$, $\min(f, g)$ measurable; in particular $f^+ := \max(f, 0)$ and $f^- := -\min(f, 0)$ are measurable. (ii) A pointwise limit of measurable functions is measurable.

Proof. (i) is Theorem 1.35 for the sequence $f, g, g, \ldots$.

(ii) If $f_n \to f$ pointwise, then $f = \limsup_n f_n$. ■

Theorem 1.37

Let $f, g : X \to \mathbb{R}$ be measurable and $F : \mathbb{R}^2 \to \mathbb{R}$ continuous. Then $h(x) := F(f(x), g(x))$ is measurable; in particular $f + g$ and fg are measurable.

Proof. Fix α. $G_\alpha := \{(u, v) \mid F(u, v) > \alpha\}$ is open in $\mathbb{R}^2$, hence (Remark 1.27(i)) a countable union of open rectangles $I_n = (\alpha_n, \beta_n) \times (\gamma_n, \delta_n)$. Each $\{(f, g) \in I_n\} = \{\alpha_n < f < \beta_n\} \cap \{\gamma_n < g < \delta_n\} \in \mathcal{M}$, so $\{h > \alpha\} = \bigcup_n \{(f, g) \in I_n\} \in \mathcal{M}$. ■

Remark 1.38

Compositions of Lebesgue measurable functions need not be Lebesgue measurable. There is a continuous increasing $f : [0, 1] \to [0, 1]$ and a Lebesgue measurable L with $f^{-1}(L)$ not Lebesgue measurable (Dudley, Prop. 4.2.1); then $g = \mathbb{1}_L$ gives $(g \circ f)^{-1}(\{1\}) = f^{-1}(L) \notin \mathcal{M}(\lambda)$. Compositions of Borel measurable functions are Borel measurable, and more generally $g \circ f$ is Lebesgue measurable if g is Borel and f Lebesgue measurable. Moreover, every Lebesgue measurable f agrees a.e. (Definition 1.49) with a Borel measurable function.

Definition 1.39 (Simple functions)

$f : X \to \mathbb{R}$ is **simple** if $\mathrm{im}(f) = \{f(x) \mid x \in X\}$ is finite. If $\mathrm{im}(f) = \{\alpha_1, \ldots, \alpha_n\}$ with distinct α_i and $E_i := \{f = \alpha_i\}$, then

$$f = \sum_{i=1}^{n} \alpha_i \mathbb{1}_{E_i}, \qquad \mathbb{1}_E(x) := \begin{cases} 1 & x \in E \\ 0 & x \notin E \end{cases} \tag{1.11}$$

($\mathbb{1}_E$ is the **indicator** or characteristic function of E). Since $\{\mathbb{1}_E \leq \alpha\}$ is $\emptyset$, E^c or X according as $\alpha < 0$, $0 \leq \alpha < 1$, $\alpha \geq 1$, the function $\mathbb{1}_E$ is measurable iff $E \in \mathcal{M}$; and f in (1.11) is measurable iff $E_1, \ldots, E_n \in \mathcal{M}$.

Theorem 1.40

For every $f : X \to \overline{\mathbb{R}}$ there are simple f_n with $f_n(x) \to f(x)$ for all x. If f is measurable the f_n can be chosen measurable, and if $f \geq 0$ they can be chosen pointwise increasing.

Proof. For $f \geq 0$ let

$$f_n := \sum_{i=1}^{n2^n} \frac{i-1}{2^n} \mathbb{1}_{\{f \in [\frac{i-1}{2^n}, \frac{i}{2^n})\}} + n \mathbb{1}_{\{f \geq n\}}. \tag{1.12}$$

Step 1 (increasing). Passing from n to $n+1$ halves each dyadic cell and the value on each half is at least the old one; on $\{f \geq n\}$ the new value is $\geq n$.

Step 2 (convergence). If $f(x) < \infty$, then for $n > f(x)$ we have $0 \leq f(x) - f_n(x) < 2^{-n}$. If $f(x) = \infty$, then $f_n(x) = n \to \infty$.

Step 3. The sets in (1.12) are measurable when f is. For general f apply this to f^+ and f^- and take $f_n^+ - f_n^-$. ∎

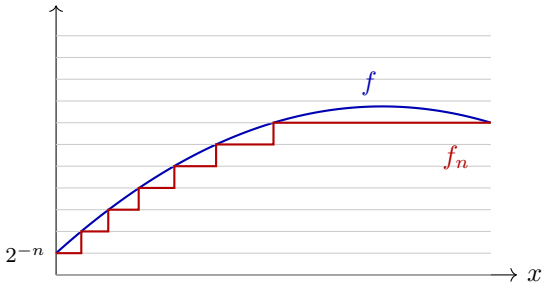


Figure. *The approximation (1.12): f_n rounds f down to the grid $2^{-n}\mathbb{Z}$ (and caps it at n). Refining the grid can only raise the step function.*

Note 1.41

If f is bounded, the sequence (1.12) for f^+ (resp. f^-) converges *uniformly*, because the error is $< 2^{-n}$ once $n > \sup |f|$. So a bounded (measurable) f is a uniform limit of (measurable) simple functions.

1.4 Integration

Motivation. The Lebesgue integral is defined by approximating f from below by simple functions, whose integral is an obvious finite sum. Its strength lies in the three limit theorems (monotone convergence, Fatou, dominated convergence), which let us exchange limits and integrals under conditions that are easy to check.

Throughout, $(X, \mathcal{M}, \mu)$ is a measure space.

Definition 1.42 (Integral of simple functions)

For $s = \sum_{i=1}^n \alpha_i \mathbb{1}_{E_i}$ with $\alpha_i > 0$, pairwise disjoint $E_i \in \mathcal{M}$, and $E \in \mathcal{M}$, put

$$I_E(s) := \sum_{i=1}^n \alpha_i \mu(E \cap E_i). \tag{1.13}$$

Lemma 1.43

If $s, \tilde{s}$ are measurable simple functions as in Definition 1.42 with $0 \le s \le \tilde{s}$, then $I_E(s) \le I_E(\tilde{s})$. In particular $I_E(s)$ does not depend on the representation of s.

Proof [Supplement]. Let $s = \sum_i \alpha_i \mathbb{1}_{E_i}$, $\tilde{s} = \sum_j \beta_j \mathbb{1}_{F_j}$. Where $s > 0$ also $\tilde{s} > 0$, so $E_i \subset \bigcup_j F_j$, and on a nonempty $E_i \cap F_j$ we have $\alpha_i \le \beta_j$. Hence, by finite additivity of μ,

$$I_E(s) = \sum_{i,j} \alpha_i \mu(E \cap E_i \cap F_j) \le \sum_{i,j} \beta_j \mu(E \cap E_i \cap F_j) \le \sum_j \beta_j \mu(E \cap F_j) = I_E(\tilde{s}).$$

For two representations of the same s, apply this in both directions. ∎

For measurable $f \ge 0$ define

$$\int_E f \, d\mu := \sup\{I_E(s) \mid s \text{ simple, measurable, } 0 \le s \le f\}. \tag{1.14}$$

Remark 1.44

$\int_E f \, d\mu \in [0, \infty]$, and $I_E(s) = I_E(s\mathbb{1}_E)$, so

$$\int_E f \, d\mu = \sup\{I_E(s) \mid s \text{ simple, measurable, } 0 \le s \le f\mathbb{1}_E\}. \tag{1.15}$$

Hence for measurable $g \ge 0$:

$$f = g \text{ on } E \Rightarrow \int_E f \, d\mu = \int_E g \, d\mu, \qquad f \le g \text{ on } E \Rightarrow \int_E f \, d\mu \le \int_E g \, d\mu, \tag{1.16}$$

and by Lemma 1.43, $\int_E s \, d\mu = I_E(s)$ for every measurable simple $s \ge 0$.

Definition 1.45 (Integral and integrability)

For measurable f with $\min\left(\int_E f^+ d\mu, \int_E f^- d\mu\right) < \infty$ put

$$\int_E f \, d\mu := \int_E f^+ d\mu - \int_E f^- d\mu \in [-\infty, \infty]. \tag{1.17}$$

If both integrals are finite, f is **(Lebesgue) integrable** on E w.r.t. μ, written $f \in \mathcal{L}(E, \mu)$ ($f \in \mathcal{L}(E)$ if $\mu = \lambda$).

Remark 1.46

For $E \in \mathcal{M}$:

(i) f measurable and bounded on E, $\mu(E) < \infty \Rightarrow f \in \mathcal{L}(E, \mu)$.

(ii) f measurable, $\alpha \le f \le \beta$ on E, $\mu(E) < \infty \Rightarrow \alpha\mu(E) \le \int_E f \, d\mu \le \beta\mu(E)$.

(iii) $f, g \in \mathcal{L}(E, \mu)$, $f \le g$ on $E \Rightarrow \int_E f \, d\mu \le \int_E g \, d\mu$.

(iv) $f \in \mathcal{L}(E, \mu) \Rightarrow \alpha f \in \mathcal{L}(E, \mu)$ and $\int_E \alpha f \, d\mu = \alpha \int_E f \, d\mu$ for $\alpha \in \mathbb{R}$.

(v) $\mu(E) = 0$, f measurable $\Rightarrow \int_E f \, d\mu = 0$.

(vi) $f \in \mathcal{L}(E, \mu)$, $A \subset E$, $A \in \mathcal{M} \Rightarrow f \in \mathcal{L}(A, \mu)$.

[Supplement] *Proofs.*
(ii) If s is simple with $0 \le s \le c\mathbb{1}_E$, then $I_E(s) \le c\mu(E)$, and the constant $c\mathbb{1}_E$ is itself admissible; hence $\int_E c \, d\mu = c\mu(E)$ for $c \ge 0$. Now $f^+ \le \beta^+$, $f^- \ge \beta^-$, $f^+ \ge \alpha^+$ and $f^- \le \alpha^-$

on E. By (1.16), $\int_E f \le \beta^+\mu(E) - \beta^-\mu(E) = \beta\mu(E)$ and similarly $\int_E f \ge \alpha\mu(E)$.
(i) follows from the bounds in (ii).
(iii) $f^+ \le g^+$ and $f^- \ge g^-$; use (1.16).
(iv) For $\alpha > 0$, $s \mapsto \alpha s$ is a bijection between simple functions below h and below αh, with $I_E(\alpha s) = \alpha I_E(s)$; so $\int_E \alpha h = \alpha \int_E h$ for $h \ge 0$, and $(\alpha f)^\pm = \alpha f^\pm$. For $\alpha < 0$ use $(\alpha f)^\pm = |\alpha| f^\mp$; $\alpha = 0$ is trivial.
(v) Every $I_E(s) = \sum \alpha_i \mu(E \cap E_i) = 0$.
(vi) $I_A(s) \le I_E(s)$ for $A \subset E$, so $\int_A f^\pm \le \int_E f^\pm < \infty$.

Theorem 1.47

(i) If $f \ge 0$ is measurable, then $\varphi(A) := \int_A f \, d\mu$, $A \in \mathcal{M}$, is σ-additive, i.e. a measure on $(X, \mathcal{M})$. (ii) The same holds for $f \in \mathcal{L}(X, \mu)$, except that φ is then a difference of two finite measures.

Proof. (ii) follows from (i) applied to f^+ and f^-. For (i) let $A = \dot{\bigcup}_i A_i$, $A_i \in \mathcal{M}$; we show $\varphi(A) = \sum_i \varphi(A_i)$.

Step 1 (simple f). For $f = \mathbb{1}_B$: $\varphi(A) = \mu(A \cap B) = \sum_i \mu(A_i \cap B) = \sum_i \varphi(A_i)$. For $f = \sum_j \beta_j \mathbb{1}_{B_j}$, $\beta_j \ge 0$, sum these identities (series of nonnegative terms may be rearranged).

Step 2 ($\le$). For simple measurable $0 \le s \le f$, Step 1 and (1.16) give $\int_A s \, d\mu = \sum_i \int_{A_i} s \, d\mu \le \sum_i \varphi(A_i)$. Taking the supremum over s: $\varphi(A) \le \sum_i \varphi(A_i)$.

Step 3 ($\ge$). φ is monotone in the set ($I_A(s) \le I_B(s)$ for $A \subset B$). So if some $\varphi(A_n) = \infty$, then $\varphi(A) = \infty$ and we are done. Otherwise, given $\varepsilon > 0$ choose simple $0 \le s_1, s_2 \le f$ with $\int_{A_k} s_k \ge \varphi(A_k) - \varepsilon$; then $s := \mathbb{1}_{A_1} s_1 + \mathbb{1}_{A_2} s_2 \le f$ and, by Step 1,

$$\varphi(A_1 \cup A_2) \ge \int_{A_1 \cup A_2} s \, d\mu = \int_{A_1} s_1 \, d\mu + \int_{A_2} s_2 \, d\mu \ge \varphi(A_1) + \varphi(A_2) - 2\varepsilon.$$

By induction $\varphi(A_1 \cup \cdots \cup A_n) \ge \sum_{i \le n} \varphi(A_i)$, and monotonicity gives $\varphi(A) \ge \sum_{i \le n} \varphi(A_i)$ for every n. ∎

Corollary 1.48

Let $A, E \in \mathcal{M}$, $A \subset E$, $\mu(E \setminus A) = 0$, and let $\int_E f \, d\mu$ be defined (e.g. $f \ge 0$ measurable or $f \in \mathcal{L}(E, \mu)$). Then $\int_E f \, d\mu = \int_A f \, d\mu$.

Proof. For $g = f^\pm$: $\int_E g = \int_A g + \int_{E \setminus A} g = \int_A g$ by Theorem 1.47 and Remark 1.46(v). ∎

Definition 1.49 (Almost everywhere)

A property P of points of $E \in \mathcal{M}$ holds **μ-almost everywhere** (μ-a.e.) on E if there is $N \in \mathcal{M}$ with $\mu(N) = 0$ such that P holds on $E \setminus N$. For Lebesgue measure one just says "a.e.".

Example 1.50

For measurable f, g: $f = g$ μ-a.e. on $E \iff \mu(\{f \ne g\} \cap E) = 0$. Here $\{f \ne g\} \in \mathcal{M}$ because $\{f < g\} = \bigcup_{q \in \mathbb{Q}} \{f < q\} \cap \{q < g\}$ and similarly for $\{f > g\}$. [Supplement]

Remark 1.51

"$f \sim g$ on $E :\iff f = g$ μ-a.e. on E" is an equivalence relation on measurable functions; for transitivity note that the union of two null sets is null. If $f \sim g$ on E and $\int_E f\,d\mu$ is defined, take a null set N with $f = g$ on $E \setminus N$. Then Corollary 1.48 and (1.16) give

$$\int_E f\,d\mu = \int_{E\setminus N} f\,d\mu = \int_{E\setminus N} g\,d\mu = \int_E g\,d\mu.$$

Null sets are negligible for integration.

Theorem 1.52

Let $E \in \mathcal{M}$.

(i) If f is measurable, $g \in \mathcal{L}(E,\mu)$ and $|f| \le g$ μ-a.e. on E, then $f \in \mathcal{L}(E,\mu)$.

(ii) $|f| \in \mathcal{L}(E,\mu)$ (i.e. $\int_E |f|\,d\mu < \infty$) $\iff$ $f \in \mathcal{L}(E,\mu)$.

(iii) $f \in \mathcal{L}(E,\mu) \Rightarrow |f| < \infty$ μ-a.e. on E.

(iv) $f \in \mathcal{L}(E,\mu) \Rightarrow |\int_E f\,d\mu| \le \int_E |f|\,d\mu$.

Proof. (i) With a null set N such that $|f| \le g$ on $E \setminus N$: $\int_E |f| = \int_{E\setminus N} |f| \le \int_{E\setminus N} g = \int_E g < \infty$. Since $f^{\pm} \le |f|$, (1.16) gives $f^{\pm} \in \mathcal{L}(E,\mu)$.

(ii) "$\Rightarrow$" by (i). "$\Leftarrow$": with $A := E \cap \{f \ge 0\}$, $B := E \cap \{f < 0\}$, $\int_E |f| = \int_A f^+ + \int_B f^- < \infty$ by Theorem 1.47 and Remark 1.46(vi).

(iii) For every n, $0 \le n\mathbb{1}_{\{|f|=\infty\}} \le |f|$, so $n\,\mu(\{|f| = \infty\} \cap E) \le \int_E |f| < \infty$; let $n \to \infty$.

(iv) By Remark 1.46(iii),(iv): $\pm \int_E f = \int_E (\pm f) \le \int_E |f|$. ∎

Theorem 1.53 (Monotone convergence theorem, Beppo Levi)

Let $0 \le f_1 \le f_2 \le \dots$ be measurable and $f := \lim_n f_n$. Then $\int_E f\,d\mu = \lim_n \int_E f_n\,d\mu$ for all $E \in \mathcal{M}$.

Proof. By (1.16), $\int_E f_n$ increases to some $\alpha \le \int_E f$. For the reverse inequality fix a simple measurable $0 \le s \le f$ and $c \in (0,1)$, and let $E_n := E \cap \{f_n \ge cs\}$. Then $E_n \nearrow E$ (if $s(x) > 0$ then $f_n(x) \to f(x) \ge s(x) > cs(x)$), and

$$\int_E f_n\,d\mu \ge \int_{E_n} f_n\,d\mu \ge c\int_{E_n} s\,d\mu.$$

Since $A \mapsto \int_A s\,d\mu$ is a measure (Theorem 1.47), continuity from below gives $\alpha \ge c\int_E s\,d\mu$. Let $c \uparrow 1$ and take the supremum over s. ∎

Theorem 1.54 (Linearity)

Let $f_1, f_2 \in \mathcal{L}(E,\mu)$ with $f_1 + f_2$ well defined (no $\infty - \infty$). Then:

(i) $f_1 + f_2$ is measurable;

(ii) if f is measurable and $f = f_1 + f_2$ μ-a.e. on E, then $f \in \mathcal{L}(E,\mu)$ and

$$\int_E f\,d\mu = \int_E f_1\,d\mu + \int_E f_2\,d\mu. \tag{1.18}$$

(1.18) also holds if $f_1, f_2 \ge 0$ are merely measurable and f is measurable with $f = f_1 + f_2$ μ-a.e. on E.

Remark 1.55

The measurability of f in Theorem 1.54(ii) is automatic if $(X, \mathcal{M}, \mu)$ is **complete**, i.e. every subset of a null set is measurable. In a complete space, g measurable and $h = g$ μ-a.e. imply h measurable. [Supplement] *Reason:* if $h = g$ off the null set N, then $\{h > \alpha\} = (\{g > \alpha\} \setminus N) \cup (\{h > \alpha\} \cap N)$, and the second set is a subset of N.

Proof of Theorem 1.54. (i) $f_1 + f_2 = f_1^+ - f_1^- + f_2^+ - f_2^-$ is the pointwise limit of the corresponding combinations of the simple approximations from Theorem 1.40, hence measurable by Corollary 1.36(ii).

(ii) By Remark 1.51 it suffices to treat $f = f_1 + f_2$.

Step 1 ($f_1, f_2 \geq 0$). For simple functions (1.18) is immediate from (1.13). In general take simple $s_n^i \nearrow f_i$ (Theorem 1.40); then $s_n^1 + s_n^2 \nearrow f_1 + f_2$ and the MCT gives $\int_E (f_1 + f_2) = \lim_n \int_E (s_n^1 + s_n^2) = \int_E f_1 + \int_E f_2$.

Step 2 (general). By Theorem 1.52(iii) and Remark 1.51 we may assume f_1, f_2 finite. Then $|f_1 + f_2| \leq |f_1| + |f_2|$, which is integrable by Step 1, so $f \in \mathcal{L}(E, \mu)$ by Theorem 1.52(i). From $f^+ - f^- = f_1^+ - f_1^- + f_2^+ - f_2^-$ we get $f^+ + f_1^- + f_2^- = f_1^+ + f_2^+ + f^-$. Step 1 gives equal integrals of both sides, all finite, and rearranging yields (1.18). ∎

Theorem 1.56

For measurable $f_k \geq 0$ and $E \in \mathcal{M}$: $\int_E \sum_{k=1}^{\infty} f_k \, d\mu = \sum_{k=1}^{\infty} \int_E f_k \, d\mu$.

Proof. $\sum_{k \leq n} f_k \nearrow \sum_k f_k$; apply the MCT and Theorem 1.54 to the partial sums. ∎

Theorem 1.57 (Fatou's lemma)

For measurable $f_n \geq 0$ and $E \in \mathcal{M}$: $\int_E \liminf_n f_n \, d\mu \leq \liminf_n \int_E f_n \, d\mu$.

Proof. $g_n := \inf_{k \geq n} f_k$ is measurable, $g_n \nearrow \liminf_n f_n$, and $g_n \leq f_k$ for $k \geq n$. By the MCT and (1.16),

$$\int_E \liminf_n f_n \, d\mu = \lim_n \int_E g_n \, d\mu \leq \liminf_n \int_E f_k \, d\mu = \liminf_n \int_E f_n \, d\mu.$$

∎

Theorem 1.58 (Dominated convergence theorem)

Let f_n be measurable with $f_n(x) \to f(x)$ for all $x \in E$, and suppose there is $g \in \mathcal{L}(E, \mu)$ (an *integrable majorant*) with

$$|f_n(x)| \leq g(x) \qquad \forall n \geq 1, \ x \in E. \tag{1.19}$$

Then $f \in \mathcal{L}(E, \mu)$ and $\int_E f \, d\mu = \lim_n \int_E f_n \, d\mu$.

Remark 1.59

Pointwise convergence in the MCT and DCT may be weakened to μ-a.e. convergence, provided the a.e. limit is measurable. That is automatic in complete spaces (Remark 1.55) but not in general. Likewise (1.19) need only hold μ-a.e., and Fatou's lemma holds for $f_n \geq 0$ μ-a.e. [Supplement] *Reason:* let N be the (countable) union of the exceptional null sets. Replacing every function by its product with $\mathbb{1}_{N^c}$ changes no integral (Corollary 1.48), and the hypotheses then hold everywhere.

Proof of Theorem 1.58. $f, f_n \in \mathcal{L}(E,\mu)$ by Theorem 1.52(i), since $|f| \le g$ as well.

Step 1. $f_n + g \ge 0$ and $f + g = \liminf_n (f_n + g)$. Fatou and Theorem 1.54 give $\int_E f + \int_E g \le \liminf_n \int_E f_n + \int_E g$, i.e. $\int_E f \le \liminf_n \int_E f_n$.

Step 2. $g - f_n \ge 0$ and $g - f = \liminf_n (g - f_n)$, so $\int_E g - \int_E f \le \int_E g + \liminf_n \int_E (-f_n)$, i.e. $\int_E f \ge \limsup_n \int_E f_n$.

Step 3. Hence $\liminf_n \int_E f_n = \limsup_n \int_E f_n = \int_E f$. ∎

1.4.1 Comparison of the Lebesgue and Riemann integrals

Motivation. The Lebesgue integral should not change the value of any integral computed in calculus. It does not, and it also characterises Riemann integrability: a bounded function is Riemann integrable exactly when its discontinuities form a null set.

Let $X = [a,b]$, $\mu = \lambda$ on $\mathcal{M} = \{A \cap [a,b] \mid A \in \mathcal{M}(\lambda)\}$; this space is complete. Write $\int_a^b f\,dx := \int_X f\,d\lambda$, $f \in \mathcal{R}([a,b])$ if f is Riemann integrable, $\mathcal{R}\int_a^b f\,dx$ for the Riemann integral and $\underline{\mathcal{R}}\int$, $\overline{\mathcal{R}}\int$ for the lower and upper Riemann integrals. For a partition $P = \{x_0, \dots, x_n\}$, $\mathrm{mesh}(P) := \max_i |x_i - x_{i-1}|$.

Theorem 1.60

(i) $f \in \mathcal{R}([a,b]) \Rightarrow f \in \mathcal{L}([a,b])$ and $\int_a^b f\,dx = \mathcal{R}\int_a^b f\,dx$.

(ii) For bounded f: $f \in \mathcal{R}([a,b]) \iff f$ is continuous a.e. on $[a,b]$.

Proof.

Step 1 (step functions). Let f be bounded (Riemann integrable functions are bounded by definition). Choose partitions $P_k \subset P_{k+1}$ with $\mathrm{mesh}(P_k) \le 1/k$ such that the lower and upper sums satisfy

$$L(P_k, f) \to \underline{\mathcal{R}}\int_a^b f\,dx, \qquad U(P_k, f) \to \overline{\mathcal{R}}\int_a^b f\,dx. \tag{1.20}$$

For $P_k = \{x_0, \dots, x_{n(k)}\}$ define step functions L_k, U_k by $L_k(a) = U_k(a) = f(a)$ and, on $(x_{i-1}, x_i]$, $L_k := m_i = \inf_{[x_{i-1},x_i]} f$, $U_k := M_i = \sup_{[x_{i-1},x_i]} f$. Then

$$L(P_k,f) = \int_a^b L_k\,dx, \qquad U(P_k,f) = \int_a^b U_k\,dx, \qquad L_1 \le L_2 \le \cdots \le f \le \cdots \le U_2 \le U_1. \tag{1.21}$$

Step 2 (limits). $L := \lim_k L_k$ and $U := \lim_k U_k$ exist and are measurable. $\max(|L_1|, |U_1|)$ is bounded, hence an integrable majorant, so the DCT, (1.21) and (1.20) give

$$\int_a^b L\,dx = \underline{\mathcal{R}}\int_a^b f\,dx, \qquad \int_a^b U\,dx = \overline{\mathcal{R}}\int_a^b f\,dx. \tag{1.22}$$

Step 3 (i). Since $L \le f \le U$,

$$f \in \mathcal{R}([a,b]) \iff \int_a^b L\,dx = \int_a^b U\,dx \iff L = U \text{ a.e.} \iff L = f = U \text{ a.e.}, \tag{1.23}$$

where the middle step is Lemma 1.62(i) applied to $U - L \ge 0$. By completeness (Remark 1.55) f is then measurable, and $\int_a^b f\,dx = \int_a^b L\,dx = \mathcal{R}\int_a^b f\,dx$ by Remark 1.51 and (1.22).

Step 4 (ii). Off the countable (hence null) set $\bigcup_k P_k$,

$$U(x) = L(x) \iff f \text{ is continuous at } x. \tag{1.24}$$

[Supplement] Indeed, such an x is interior to an interval (x_{i-1}, x_i) of each P_k, of length $\le 1/k$. If f is continuous at x and $\varepsilon > 0$, then for large k this interval lies in a δ-neighbourhood on which $|f - f(x)| < \varepsilon$, so $U_k(x) - L_k(x) \le 2\varepsilon$. Conversely, if $U(x) = L(x)$, choose k with $U_k(x) - L_k(x) < \varepsilon$; then $|f(y) - f(x)| \le M_i - m_i < \varepsilon$ on the open interval around x. Combining (1.23) and (1.24): $f \in \mathcal{R}([a,b]) \iff L = U$ a.e. $\iff f$ is continuous a.e. ∎

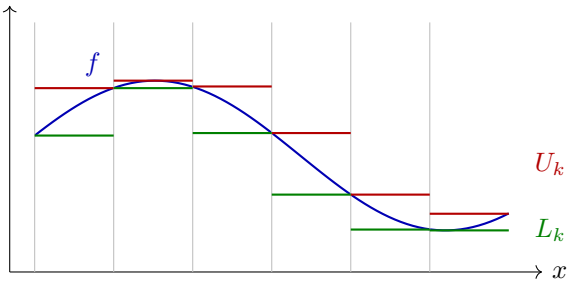


Figure. *Lower and upper step functions of a partition. Their integrals are the lower and upper sums, and refining the partition squeezes them towards f at every continuity point.*

1.5 Some fundamental inequalities and the L^p-spaces

Motivation. Integrals of powers $|f|^p$ measure the size of a function. Markov, Jensen, Hölder and Minkowski control these sizes, and Minkowski's inequality makes $\|f\|_p$ a (semi-)norm. The payoff is completeness: L^p is a Banach space, the natural home for limits of random variables.

Again $(X, \mathcal{M}, \mu)$ is an arbitrary measure space.

Proposition 1.61 (Markov ($p = 1$), Chebyshev ($p = 2$))

For measurable f, $p \in (0, \infty)$ and $\varepsilon > 0$:

$$\mu(\{|f| \ge \varepsilon\}) \le \frac{1}{\varepsilon^p} \int_X |f|^p \, d\mu.$$

Proof. $\mathbb{1}_{\{|f| \ge \varepsilon\}} = \mathbb{1}_{\{|f|^p \ge \varepsilon^p\}} \le \varepsilon^{-p} |f|^p$. Integrate, using (1.16) and Remark 1.46(iv) (whose proof for $h \ge 0$ does not need integrability). ∎

Lemma 1.62

Let f be measurable. (i) If $f \ge 0$ μ-a.e. and $\int_X f \, d\mu = 0$, then $f = 0$ μ-a.e. (ii) If $\int_A f \, d\mu = 0$ for every $A \in \mathcal{M}$, then $f = 0$ μ-a.e.

Proof [Supplement]. (i) $|f| = f$ a.e., so $\int |f| \, d\mu = 0$ by Remark 1.51. Markov gives $\mu(\{|f| \ge \frac{1}{n}\}) \le n \int |f| = 0$, and $\{f \ne 0\} = \bigcup_n \{|f| \ge \frac{1}{n}\}$.

(ii) With $A := \{f > 0\}$, $f \mathbb{1}_A \ge 0$ and $\int_X f \mathbb{1}_A \, d\mu = \int_A f \, d\mu = 0$, so $f \mathbb{1}_A = 0$ a.e. by (i); since $f \mathbb{1}_A > 0$ on A, $\mu(A) = 0$. Likewise $\mu(\{f < 0\}) = 0$. ∎

Theorem 1.63 (Jensen's inequality)

Let $(X, \mathcal{M}, \mu)$ be a probability space, $f \in \mathcal{L}(X, \mu)$ real-valued with $f(x) \in (a, b)$ for all x, and

$\varphi : (a,b) \to \mathbb{R}$ convex. Then $(\varphi \circ f)^- \in \mathcal{L}(X,\mu)$ and

$$\varphi\Big(\int_X f\,d\mu\Big) \le \int_X \varphi \circ f\,d\mu \in (-\infty,\infty].$$

Proof.

Step 1. A convex φ is continuous, so $\varphi \circ f$ is measurable. Put $t := \int_X f\,d\mu \in (a,b)$ (since $a < f < b$ and $\mu(X) = 1$). Convexity gives the supporting line at t, $\varphi(y) \ge \varphi(t) + \beta_+(y-t)$ for $y \in (a,b)$, where β_+ is the right derivative of φ at t. Hence

$$\varphi \circ f \ge g := \varphi(t) + \beta_+(f-t) \in \mathcal{L}(X,\mu). \tag{1.25}$$

Step 2. From (1.25), $(\varphi \circ f)^- \le g^-$, so $(\varphi \circ f)^- \in \mathcal{L}(X,\mu)$ and $\int \varphi \circ f$ is defined in $(-\infty,\infty]$.

Step 3. Since g is finite a.e., (1.25) reads $(\varphi \circ f)^+ + g^- \ge (\varphi \circ f)^- + g^+ \ge 0$ a.e. Integrating (monotonicity and Theorem 1.54 for nonnegative functions) and rearranging the finite terms $\int (\varphi \circ f)^-$, $\int g^-$:

$$\varphi\Big(\int_X f\,d\mu\Big) = \int_X g\,d\mu \le \int_X \varphi \circ f\,d\mu.$$

■

Example 1.64 (AM–GM)

Let $X = \{p_1,\dots,p_n\}$, $\mu(\{p_i\}) = \alpha_i > 0$ with $\sum \alpha_i = 1$, $f(p_i) = x_i$, $\varphi = \exp$. Jensen gives $e^{\sum \alpha_i x_i} \le \sum \alpha_i e^{x_i}$, i.e. with $y_i = e^{x_i}$,

$$\prod_{i=1}^n y_i^{\alpha_i} \le \sum_{i=1}^n \alpha_i y_i \qquad (\alpha_i, y_i > 0,\ \textstyle\sum \alpha_i = 1).$$

Theorem 1.65 (Hölder and Minkowski inequalities)

Let $f, g : X \to [0,\infty]$ be measurable and $p, q \in (1,\infty)$ with $\frac{1}{p} + \frac{1}{q} = 1$. Then

$$\int_X fg\,d\mu \le \Big(\int_X f^p\,d\mu\Big)^{1/p}\Big(\int_X g^q\,d\mu\Big)^{1/q}, \tag{1.26}$$

$$\Big(\int_X (f+g)^p\,d\mu\Big)^{1/p} \le \Big(\int_X f^p\,d\mu\Big)^{1/p} + \Big(\int_X g^p\,d\mu\Big)^{1/p}. \tag{1.27}$$

Proof.

Step 1 (Hölder, trivial cases). Let A, B be the two factors on the right of (1.26). If $A = 0$, then $f = 0$ a.e. (Lemma 1.62), so $fg = 0$ a.e. and (1.26) holds; similarly if $B = 0$. If $A, B > 0$ and one of them is ∞, (1.26) is trivial. So let $A, B \in (0,\infty)$, $F := f/A$, $G := g/B$; then $\int F^p = \int G^q = 1$.

Step 2 (Young). Where $F(x), G(x) \in (0,\infty)$ write $F = e^{s/p}$, $G = e^{t/q}$; convexity of exp gives $FG = e^{s/p + t/q} \le \frac{1}{p}e^s + \frac{1}{q}e^t = \frac{1}{p}F^p + \frac{1}{q}G^q$. This also holds when F or $G \in \{0,\infty\}$. Integrating: $\int FG \le \frac{1}{p} + \frac{1}{q} = 1$, which is (1.26).

Step 3 (Minkowski). $(f+g)^p = f(f+g)^{p-1} + g(f+g)^{p-1}$; apply (1.26) to each term and use $(p-1)q = p$:

$$\int (f+g)^p \le \Big[\Big(\int f^p\Big)^{1/p} + \Big(\int g^p\Big)^{1/p}\Big]\Big(\int (f+g)^p\Big)^{1/q}. \tag{1.28}$$

Step 4. We may assume $\int (f+g)^p > 0$ and $\int f^p + \int g^p < \infty$. Convexity of $t \mapsto t^p$ gives $(\frac{f+g}{2})^p \le \frac{1}{2}(f^p + g^p)$, so $C := (\int (f+g)^p)^{1/q} \in (0,\infty)$. Dividing (1.28) by C gives (1.27) since $1 - \frac{1}{q} = \frac{1}{p}$. ■

Definition 1.66 ($\mathcal{L}^p$)

For $p \in [1,\infty)$ let $\mathcal{L}^p(X,\mu) := \{f : X \to \mathbb{R} \text{ measurable} \mid \|f\|_p := \left(\int_X |f|^p \, d\mu\right)^{1/p} < \infty\}$.

Theorem 1.67

(i) For $p \in [1,\infty)$, $\mathcal{L}^p(X,\mu)$ is a vector space and $\|\cdot\|_p$ a semi-norm on it. If (f_n) is Cauchy in $\mathcal{L}^p$, there are $f \in \mathcal{L}^p(X,\mu)$ and a subsequence (f_{n_k}) with $\|f - f_n\|_p \to 0$ and $f_{n_k} \to f$ μ-a.e.

(ii) If $p,q \in (1,\infty)$, $\frac{1}{p} + \frac{1}{q} = 1$, $f \in \mathcal{L}^p$, $g \in \mathcal{L}^q$, then $fg \in \mathcal{L}^1$ and

$$\|fg\|_1 \le \|f\|_p \|g\|_q. \tag{1.29}$$

Proof.

Step 1 (semi-norm). $\|\alpha f\|_p = |\alpha| \|f\|_p$ is clear, and $|f+g| \le |f| + |g|$ with Minkowski gives $\|f+g\|_p \le \|f\|_p + \|g\|_p$ (for $p = 1$ by linearity). So $\mathcal{L}^p$ is a vector space. $\|f\|_p = 0$ only gives $|f|^p = 0$ a.e. (Lemma 1.62(i)), i.e. $f = 0$ a.e.; hence a semi-norm only.

Step 2 (a.e. convergent subsequence). Choose $n_1 < n_2 < \dots$ with $\|f_{n_{k+1}} - f_{n_k}\|_p < 2^{-k}$ and let $g_k := f_{n_{k+1}} - f_{n_k}$. By the MCT and Minkowski,

$$\Big\| \sum_k |g_k| \Big\|_p = \lim_n \Big\| \sum_{k \le n} |g_k| \Big\|_p \le \limsup_n \sum_{k \le n} \|g_k\|_p < 1.$$

By Theorem 1.52(iii) there is a null set N with $\sum_k |g_k(x)| < \infty$ off N. There $f_{n_{k+1}}(x) = f_{n_1}(x) + \sum_{l \le k} g_l(x)$ converges. Put $f := \lim_k f_{n_k} \mathbb{1}_{X \setminus N}$, which is measurable with $f_{n_k} \to f$ a.e.

Step 3 (L^p convergence). Given $\varepsilon > 0$ take N_0 with $\|f_m - f_n\|_p < \varepsilon$ for $m,n \ge N_0$. For $n_k \ge N_0$, Fatou gives $\|f - f_{n_k}\|_p \le \liminf_l \|f_{n_l} - f_{n_k}\|_p \le \varepsilon$. Hence $f = (f - f_{n_k}) + f_{n_k} \in \mathcal{L}^p$ and $f_{n_k} \to f$ in $\mathcal{L}^p$; a Cauchy sequence with a convergent subsequence converges, so $f_n \to f$ in $\mathcal{L}^p$.

(ii) is (1.26) applied to $|f|, |g|$. ∎

Remark 1.68

(i) Identifying functions that agree μ-a.e. turns $\mathcal{L}^p$ into a normed space: $L^p(X,\mu) := \mathcal{L}^p(X,\mu)/\|\cdot\|_p^{-1}(0)$, where $\|\cdot\|_p^{-1}(0) = \{f \text{ measurable} \mid f = 0 \ \mu\text{-a.e.}\}$. For the class $\tilde{f}$ of f put

$$\|\tilde{f}\|_p := \|f\|_p; \tag{1.30}$$

now $\|\tilde{f}\|_p = 0 \iff f = 0$ a.e. $\iff \tilde{f} = 0$. In particular $L^2(X,\mu)$ is an inner product space with

$$\langle \tilde{f}, \tilde{g} \rangle_2 := \int_X fg \, d\mu, \qquad \|\tilde{f}\|_2 = \sqrt{\langle \tilde{f}, \tilde{f} \rangle_2}, \tag{1.31}$$

for arbitrary representatives f, g.

(ii) In practice we work with representatives and keep in mind the identification $f \sim g \iff f = g$ μ-a.e.

(iii) $\mathcal{L}^p$ uses real-valued functions so that sums are always defined. A measurable $g : X \to [0,\infty]$ with $\int g^p < \infty$ is finite a.e. (Theorem 1.52(iii)) and is tacitly replaced by a finite version.

Example 1.69

For $X = \mathbb{N}$, $\mathcal{M} = \mathcal{P}(\mathbb{N})$, $\mu = \sum_k \delta_k$ (counting measure), $\emptyset$ is the only null set and

$$L^p(X,\mu) = \mathcal{L}^p(X,\mu) = \ell^p := \Big\{ x = (x_k) \in \mathbb{R}^{\mathbb{N}} \;\Big|\; \sum_k |x_k|^p < \infty \Big\}.$$

Theorem 1.70 (Riesz–Fischer)

For $p \in [1,\infty)$, $(L^p(X,\mu), \|\cdot\|_p)$ is a Banach space; in particular $L^2(X,\mu)$ is a Hilbert space with the inner product (1.31).

Proof. By Remark 1.68(i) it is a normed space (an inner product space for $p = 2$). If $(\tilde{f}_n)$ is Cauchy, the representatives (f_n) are Cauchy in $\mathcal{L}^p$ by (1.30). Theorem 1.67(i) gives $f \in \mathcal{L}^p$ with $\|f - f_n\|_p \to 0$, i.e. $\|\tilde{f} - \tilde{f}_n\|_p \to 0$. ■